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PROBABILITY

Probability can be defined as the possibility of occurrence of an event. Probability is the likelihood or the chances that an uncertain event will occur. The probability of an event always lies between 0 and 1.

$$\text{Probability } (P) = \frac{\text{Favourable Outcomes}}{\text{Total Outcomes}}$$

$$P(A) = \frac{n(A)}{n(S)}$$

Where, $P(A)$ represents the probability of event A,

$n(A)$ represents the number of favorable outcomes, and

$n(S)$ represents total outcomes in sample space.

I. Theorem of Complementary Events

Two events are said to be complementary events if the sum of their probability is one. Thus, if A is an event and the probability of A is given by P(A) then this theorem states that,

$$P(A') = 1 - P(A)$$

where $P(A')$ is the probability of the complementary event of A; i.e., A'. In such cases, events A and A' are said to be mutually exhaustive also.

II. Theorem of Addition

If A, B, C are not mutually exclusive Events:

1. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ OR $P(A \; or \; B) = P(A) + P(B) - P(A\&B)$
2. $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

If A, B, C are mutually exclusive events, then $P(A \cap B) = 0, P(B \cap C) = 0, P(C \cap A) = 0$

$P(A \cap B \cap C) = 0$

1. $P(A \cup B) = P(A) + P(B)$
2. $P(A \cup B \cup C) = P(A) + P(B) + P(C)$

Example:

If A and B are events of occurrence of 2 and occurrence of 3 on a dice respectively, then calculate the probability of occurrence of A or B; i.e., getting 2 or 3 on dice.

Solution: Probability of getting 2: $P(A) = 1/6$ and Probability of getting 3: $P(B) = 1/6$

As A and B are independent and mutually exclusive events, using the theorem of addition,

$$P(A \cup B) = P(A) + P(B)$$

$$P(A \cup B) = 1/6 + 1/6$$

$$P(A \cup B) = 1/3.$$

2. When the Events are Overlapping.

When the events are overlapping, the theorem of addition determines the probability that one or more events would occur in a single trial.

If E_1 and E_2 are overlapping events, then according to this theorem:

$$P(E1 \cup E2) = P(E1) + P(E2) - P(E1 \cap E2)$$

where, $P(E1 \cup E2)$ is the probability that either E_1 or E_2 or both events will occur, and $P(E_1 \cap E_2)$ is the joint probability that indicates the probability of occurrence of both E_1 and E_2.

If there are three events E_1, E_2, and E_3, then

$$P(E1 \cup E2 \cup E3) = P(E1) + P(E2) + P(E3) - P(E1 \cap E2) - P(E1 \cap E3) - P(E2 \cap E3) + P(E1 \cap E2 \cap E3)$$

where $P(E_1 \cup E_2 \cup E_3)$ gives the probability of occurrence of at least one of the events; E_1, E_2, and E_3.

Example:

Consider a class with 20 students. 10 students passed in Math's, 15 in English, and 13 in both. What is the probability that a student passed in either Math or English?

Solution: Total students = 20

P(A) = Probability that a student passed in Math's = $10/20$

P(B) = Probability that a student passed in English = $15/20$

$P(A \cap B)$ = Probability that a student passed in both Math and English = $13/20$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cup B) = 3/5.$

III. Theorem of Multiplication

1. When events are Independent

If A and B are independent events with $P(A) \neq 0$ and $P(B) \neq 0$, then according to this theorem of probability:

$$P(A \cap B) = P(A)P(B) \text{ and } P(\bar{A} \cap \bar{B}) = P(\bar{A})P(\bar{B})$$

This means that the probability of intersection of two events A and B is equal to the product of the individual probabilities of events A and B.

If there are more than two events; say, A, B and C then,

$$P(A \cap B \cap C) = P(A)P(B)P(C) \text{ and } P(\bar{A} \cap \bar{B} \cap \bar{C}) = P(\bar{A})P(\bar{B})P(\bar{C})$$

Example:

1. The probability that A can solve the problem is $\frac{2}{3}$ and B can solve it is $\frac{3}{4}$. If both attempt the

problem, then the probability that the problem get solved.

Answer - Let $\mathrm{P(A)} = \frac{2}{3}$, $\mathrm{P(B)} = \frac{3}{4}$ and $\mathrm{P}(\overline{\mathrm{A}}) = 1 - \mathrm{P(A)} = 1 - \frac{2}{3} = \frac{1}{3}$

$$\mathrm{P}(\overline{\mathrm{B}}) = 1 - \mathrm{P(B)} = 1 - \frac{3}{4} = \frac{1}{4}$$

Required probability $\mathrm{P} = \mathrm{P}(\mathrm{A} \cap \mathrm{B}) + \mathrm{P}(\mathrm{A} \cap \overline{\mathrm{B}}) + \mathrm{P}(\overline{\mathrm{A}} \cap \mathrm{B})$

$\mathrm{P} = \mathrm{P(A)P(B)} + \mathrm{P(A)P}(\overline{\mathrm{B}}) + \mathrm{P}(\overline{\mathrm{A}})\mathrm{P(B)}$

$$\mathrm{P} = \frac{2}{3} * \frac{3}{4} + \frac{2}{3} * \frac{1}{4} + \frac{1}{3} * \frac{3}{4} = \frac{11}{12}.$$

2. A statistics problem is given to two students, say A and B. Their chances of solving it correctly are known to be 0.5 and 0.3, respectively. Find the probability that both of them solve it.

Solution: Let E_1 be the event of A solving the problem and E_2 is the event of B solving the problem. Here event E_1 and E_2 are independent.

$$P(E1) = 0.5$$

$$P(E2) = 0.3$$

$$P(E1 \cap E2) = P(E1) \times P(E2)$$

$$P(E1 \cap E2) = 0.5 \times 0.3$$

$$P(E1 \cap E2) = 0.15$$

Therefore, the chances of both A and B solving the problem is 15%.

2. When Events are not Independent

If the events are not independent, then multiplication theorem states that the joint probability of the events E_1 and E_2 is given by the probability of event E_1 multiplied by the probability of event E_2 given that event E_1 has occurred and vice-versa. Simply put, the rule uses the concept of conditional probability when the events are known to be dependent or non-independent. According to this theorem, if E_1 and E_2 are two events where $P(E_1) \neq 0$ and $P(E_2) \neq 0$, and if E_1 and E_2 are not independent events, then:

$$P(E1 \cap E2) = P(E1).P(E2/E1)$$

$$P(E1 \cap E2) = P(E2).P(E1/E2)$$

Similarly, if there are three dependent events E_1, E_2, and E_3, then,

$$P(E1 \cap E2 \cap E3) = P(E1).P(E2/E1).P(E3/E1 \cap E2)$$

Example:

1. In a class 40% students read statistics, 25% read mathematics and 15% read both statistics and mathematics. One student is selected at random. The probability that, he read statistics if he knows that he read mathematics.

Solution: Let A-be the event that students read statistics $P(A) = \frac{40}{100} = 0.40$

B- be the event that students read Mathematics $P(B) = \frac{25}{100} = 0.25$

both read stat and math's $P(A \cap B) = \frac{15}{100} = 0.15$

Here find P(AIB) by $P(AIB) = \frac{P(A \cap B)}{P(B)} = \frac{0.15}{0.25} = \frac{3}{5} = 0.6.$

2. A large company employs 70 engineers, of whom 36 are males and the remaining are females. Of the female engineers. 14 are under 35 years of age, 15 are between 35 and 45 years of age, and the remaining are over 45 years of age. What is the probability of randomly selected engineer who is a female and under the age of 35 years of age?

Solution: Let E_1 represent the event that an engineer selected at random is a female and E_2 is the event that a female engineer selected is under 35 years of age.

Since there are 36 males out of 70 engineers, it means that the number of female engineers is 34.

$$P(E1) = \frac{Total\ Female\ Engineers}{Total\ Engineers}$$

$$P(E1) = \frac{34}{70}$$

$$P\left(\frac{E2}{E1}\right) = \frac{14}{34}$$

$$\text{Therefore, } P(E1 \cap E2) = P(E1) \times P\left(\frac{E2}{E1}\right)$$

$$P(E1 \cap E2) = \frac{34}{70} \times \frac{14}{34}$$

$$P(E1 \cap E2) = \frac{1}{5}.$$

RANDOM VARIABLE

A random variable: It is a real valued function that assigns a real number to each outcome in the sample space of a random experiment.

Example1: A coin is tossed which results in head or tail. Let X is the random variable whose value for any outcome is the number of heads obtained. Find the probability function of X and construct a probability distribution table.

Answer: Let H-head, T-tail then sample space is, $S = \{H, T\}$, Total number of sample space is 2. X(H) = 1, $X(T)$ = 1 and x is the number of heads which takes the value 0 and 1.

$P(x = 0) = 1/2$ and $P(x = 1) = 1/2$ Probability distribution table is:

Random Variable X:	X=0	X=1
P(x)	1/2	1/2

Example2: A coin is tossed two times successively. Let X is the random variable whose value for any outcome is the number of heads obtained. Find the probability function of X and construct a probability distribution table.

Answer: All possible outcomes are $2^2 = 4$, and sample space is, $S = \{HH, TT, HT, TH\}$

x is number of heads which takes the value 0, 1, 2.

$X(x = 0) = 1, X(x = 1) = 2, X(x = 2) = 1$

i. No head $\Rightarrow P(X = 0) = P(TT) = \frac{1}{4}$

ii. One head $\Rightarrow P(X=1) = P(HT) + p(TH) = \frac{2}{4}$

iii. Two head $\Rightarrow P(X=2) = P(HH) = \frac{1}{4}$

Probability distribution table is

x	0	1	2
$P(x)$	$1/4$	$2/4$	$1/4$

Example3: For an experiment of simultaneous throw with three coins. The results of three tosses are independent of each other. Find the probability function and distribution table.

Answer: All possible outcomes are $2^3 = 8$, and sample space is,

$S = \{HHH, TTT, HHT, HTH, HTT, TTH, THT, THH\}$.

Let x denotes the number of heads appearing in each case.

i. Probability of No head $\Rightarrow P(x=0) = 1/8$.

ii. Probability of 1 head $\Rightarrow P(x=1) = 3/8$.

iii. Probability of 2 heads $\Rightarrow P(x=2) = 3/8$.

iv. Probability of 3 heads $\Rightarrow P(x=3) = 1/8$

Probability distribution table is

x	0	1	2	3
$P(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Example: Suppose a die is thrown (X = outcome of the dice) once.

Sample space S = {1, 2, 3, 4, 5, 6}. The output of the function will be:

Random Variable	X=1	X=2	X=3	X=4	X=5	X=6
Probability	1/6	1/6	1/6	1/6	1/6	1/6

Types of Random Variables: (i) Discrete Random Variable (ii) Continuous Random Variable

DISCRETE RANDOM VARIABLE

A variable which can assume only a countable number of real values and for which the value which the variable takes depends on a chance, is called a d**iscrete random**

variable. In other words, a real valued function defined on a discrete sample space is called a discrete random variable.

Examples of discrete random variables are marks obtained in a test, number of accidents per months, number of telephone calls per minute, number of success in n trials, and so on…

Probability Mass Function (PMF):

Definition: If X is a discrete random variable such that $P(X = x) = P_i$ then P_i is said to be probability mass function (p.m.f.) if it satisfy following conditions

1. $P_i \geq 0$.
2. $\sum_{i=0}^{n} P(x_i) = 1$.

MATHEMATICAL EXPECTATION, MEAN AND VARIANCE

If X is random variable and probability function ∅(x) be any function of x then expectation of ∅(x) is denoted by $E\big(\emptyset(x)\big)$ and defined as

1. $E\big(\emptyset(x_i)\big) = \sum_{i=1}^{n} \emptyset(x_i)P(x_i)$ … $(In\ Discrete\ Random\ Variable)$.
2. $E\big(\emptyset(x_i)\big) = \int_{-\infty}^{\infty} \emptyset(x_i)f(x)\,dx$ … $(In\ Continuous\ Random\ Variable)$.

If $\emptyset(x_i) = x$ then **Mean** $= E(x)$ is

1. $E(x) = \sum_{i=1}^{n} x_i\, P(x_i)$ … $(In\ Discrete\ Random\ Variable)$.
2. $E(x) = \int_{-\infty}^{\infty} x\, f(x)\,dx$ … $(In\ Continuous\ Random\ Variable)$.

If $\emptyset(x_i) = x$ then **Variance** $= E(x - \bar{x})^2$ in Discrete Random Variable is defined as

$v(x) = E(x - \bar{x})^2 = E(x^2 - 2x\bar{x} + \bar{x}^2)$

$v(x) = E(x^2) - 2\bar{x}E(x) + E(\bar{x}^2) = E(x^2) - 2\bar{x}\,\bar{x} + (\bar{x}^2)$

$v(x) = E(x^2) - (\bar{x}^2)$

$v(x) = E(x^2) - \big(E(x)\big)^2$

Examples*:*

1. A random variable X has the following probability distribution

x	0	1	2	3	4	5	6	7
$P(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

Find $i)\ k\quad ii)\ P(x<6)\quad iii)\ P(x\geq 6)\quad iv)\ P(0<x<5)$.

Answer: We know that sum of all probabilities is one. i.e. $\sum_{x=0}^{7} P(x)=1$

$\therefore k+2k+2k+3k+k^2+2k^2+7k^2+k=1$

$10k^2+9k=1$

$10k^2+9k-1=0\rightarrow k=-1\ or\ \frac{1}{10}$

k $=-1$ is rejected as probability cannot be negative. so, use, k $=\frac{1}{10}$

ii) P(x <6) $=$ P(0) $+$ P(1) $+$ P(2) $+\cdots+$ P(5) $=0+\frac{1}{10}+\frac{2}{10}+\frac{2}{10}+\frac{3}{10}+\frac{1}{100}=\frac{81}{100}$.

iii) P(x ≥ 6) $=$ P(6) $+$ P(7) $=2\left(\frac{1}{10}\right)^2+7\left(\frac{1}{10}\right)^2+\frac{1}{10}=\frac{19}{100}$.

iii) P(0 $<$ x <5) $=$ P(1) $+$ P(2) $+$ P(3) $+$ P(4) $=\frac{1}{10}+\frac{2}{10}+\frac{2}{10}+\frac{3}{10}=\frac{8}{10}$.

2. The outcomes of a certain experiment are $x_1=1, x_2=2, x_3=3$. The associated probability

function is $P(x)=\begin{cases} k & x=1 \\ 2k & x=2 \\ 5k & x=3 \end{cases}$ Find $P(x<2), P(x\leq 2), P(x\leq 3)$.

Answer: We know that sum of all probabilities is one. i.e. $\sum_{x=1}^{3} P(x)=1$

$\therefore k+2k+5k=1\rightarrow k=\frac{1}{8}$

$\therefore P(x=1)=k=\frac{1}{8}$

$\therefore P(x=2)=2k=\frac{2}{8}$

$\therefore P(x=3)=5k=\frac{5}{8}$

(i). $P(x<2)=P(x=1)=\frac{1}{8}$.

(ii). $P(x\leq 2)=P(=1)+P(x=2)=\frac{1}{8}+\frac{2}{8}=\frac{3}{8}$.

(iii). $P(x\leq 3)=P(=1)+P(x=2)+P(x=3)=\frac{1}{8}+\frac{2}{8}+\frac{5}{8}=\frac{8}{8}=1$.

3. Given the following probability function,

x	0	1	2	3	4	5
$P(x)$	0	c	2c	2c	c	$7c^2$

Find $i)\ c\quad ii)\ P(x \geq 2)\quad iii)\ P(x < 3)$.

Answer: (i) We know that sum of all probabilities is one.

$c + 2c + 2c + c + 7c^2 = 1$

$7c^2 + 6c = 1 \rightarrow 7c^2 + 6c - 1 = 0$

$(7c - 1)(c + 1) = 0$

$c = -1\ or\ c = \frac{1}{7}$

$c = -1$ is rejected as probability cannot be negative. so, use $c = \frac{1}{7}$

ii) $P(x \geq 2) = P(x = 2) + P(x = 3) + P(x = 4) + P(x = 5)$

$\therefore P(x \geq 2) = 2c + 2c + c + 7c^2$

$\therefore P(x \geq 2) = 5c + 7c^2 = \frac{5}{7} + \frac{7}{49} = \frac{6}{7}.$

iii) $P(x < 3) = P(x = 0) + P(x = 1) + P(x = 2)$

$\therefore P(x < 3) = 0 + c + 2c = 3c = \frac{3}{7}.$

4. A discrete random variable X has a probability mass function as $P(x) = cx$ for $x = 0,1,2,3,4,5$. Find (i) c (ii) $P(X \leq 4)$ (iii) Mean of X.

Answer: Probability mass function is $P(x) = cx$

x	0	1	2	3	4	5
$P(x)$	0	c	2c	3c	4c	5c

We know that sum of all probabilities is one $c + 2c + 3c + 4c + 5c = 1 \rightarrow c = \frac{1}{15}$

$P(x \leq 4) = 0 + c + 2c + 3c + 4c = 10c = \frac{10}{15} = \frac{2}{3}.$

Mean $E(x) = \sum x\, P(x) = c + 4c + 9c + 16c + 25c = 55c = \frac{55}{15} = \frac{11}{3}.$

5. Let X be a discrete random variable with the following PMF $P(x) = \begin{cases} 0.1 & for\ x = 0.2 \\ 0.2 & for\ x = 0.4 \\ 0.2 & for\ x = 0.5 \\ 0.3 & for\ x = 0.8 \\ k & for\ x = 1 \\ 0 & otherwise \end{cases}$

   a. Find R_X, the range of the random variable X.

   b. Find k.

   c. Find $P(X \leq 0.5)$.

   d. Find $P(0.25 < X < 0.75)$.

   Answer: (a) The range of X can be found from the pmf. The range of X consists of possible values of X are $R_X = \{0.2, 0.4, 0.5, 0.8,\ 1\}$

   (b) We know that sum of all probabilities is one $0.1 + 0.2 + 0.2 + 0.3 + k = 1 \rightarrow k = 0.2$

   (c) $P(X \leq 0.5) = P(X = 0.2) + P(X = 0.4) + P(X = 0.5) = 0.1 + 0.2 + 0.2 = 0.5.$

   (d) $P(0.25 < X < 0.75) = P(X = 0.4) + P(X = 0.5) = 0.2 + 0.2 = 0.4.$

6. A die is tossed once. Random variable X, denote the digit that appears. . Find the Mean (expected value) and Variance.

   Answer: Probability distribution table is

x	1	2	3	4	5	6
$P(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

   Mean $E(x) = \sum x\, P(x) = 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + 3 \times \frac{1}{6} + 4 \times \frac{1}{6} + 5\frac{1}{6} + 6 \times \frac{1}{6} = \frac{21}{6} = \frac{7}{2}.$

   $E(x^2) = \sum x^2 P(x) = 1 \times \frac{1}{6} + 4 \times \frac{1}{6} + 9 \times \frac{1}{6} + 16 \times \frac{1}{6} + 25\frac{1}{6} + 36 \times \frac{1}{6} = 15.17.$

   Variance $= E(x^2) - \left(E(x)\right)^2 = 2.91.$

7. A die is tossed twice. X denotes the sum of digits in two throws. Find the Mean (expected value) and Variance.

   Answer: Number of sample space is 36. Probability distribution table is,

x	2	3	4	5	6	7	8	9	10	11	12
$P(x)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

   Mean $E(x) = \sum x\, P(x)$

$E(x) = 2\frac{1}{36} + 3\frac{2}{36} + 4\frac{3}{36} + 5\frac{4}{36} + 6\frac{5}{36} + 7\frac{6}{36} + 8\frac{5}{36} + 9\frac{4}{36} + 10\frac{3}{36} + 11\frac{2}{36} + 12\frac{1}{36} = 7.$

$E(x^2) = \sum x^2 P(x) = 4\frac{1}{36} + 9\frac{2}{36} + 16\frac{3}{36} + 25\frac{4}{36} + 36\frac{5}{36} + 49\frac{6}{36} + 64\frac{5}{36} + 81\frac{4}{36} +$

$100\frac{3}{36} + 121\frac{2}{36} + 144\frac{1}{36} = 54.83.$

Variance $= E(x^2) - \big(E(x)\big)^2 = 5.83.$

8. A coin is thrown 4 times successively. X denotes the number of heads. Find the Mean and Variance.

Answer: All possible outcomes are $2^4 = 16$, and sample space is,

Sample Space S = {HHHH, TTTT, HTHH, HTTH, HTTT, THHH, THHT, HHTT, TTHH, TTTH, THTH, HTHT, HHHT, THTT, TTHT, HHTH}

$n(S) = 16$

i. Probability of No head ⇒ $P(x = 0) = 1/16$

ii. Probability of 1 head ⇒ $P(x = 1) = 4/16$

iii. Probability of 2 heads ⇒ $P(x = 2) = 6/16$

iv. Probability of 3 heads ⇒ $P(x = 3) = 4/16$

v. Probability of 4 heads ⇒ $P(x = 4) = 1/16$

Probability distribution table is,

x	0	1	2	3	4
$P(x)$	1/16	4/16	6/16	4/16	1/16

Mean $E(x) = \sum x\,P(x) = 0 \times \frac{1}{16} + 1 \times \frac{4}{16} + 2 \times \frac{6}{16} + 3 \times \frac{4}{16} + 4 \times \frac{1}{16} = \frac{32}{16} = 2.$

$E(x^2) = \sum x^2\,P(x) = 0 \times \frac{1}{16} + 1 \times \frac{4}{16} + 4 \times \frac{6}{16} + 9 \times \frac{4}{16} + 16 \times \frac{1}{16} = 5.$

Variance $= E(x^2) - \big(E(x)\big)^2 = 1.$

9. Variable x takes the values 0, 1, 2, 3, 4, 5 with probability of each as $\frac{1}{15}, \frac{2}{15}, \frac{3}{15}, \frac{3}{15}, \frac{5}{15}, \frac{1}{15}$.

Find the Mean and Variance.

Answer: Probability distribution table is

x	0	1	2	3	4	5
P(x)	1/15	2/15	3/15	3/15	5/15	1/15

$$E(x) = \sum x\,P(x) = 0 \times \frac{1}{15} + 1 \times \frac{2}{15} + 2 \times \frac{3}{15} + 3 \times \frac{3}{15} + 4 \times \frac{5}{15} + 5 \times \frac{1}{15} = 2.8.$$

$$E(x^2) = \sum x^2\,P(x) = 0 \times \frac{1}{15} + 1 \times \frac{2}{15} + 4 \times \frac{3}{15} + 9 \times \frac{3}{15} + 16 \times \frac{5}{15} + 25 \times \frac{1}{15} = 9.73.$$

$$\text{Variance} = E(x^2) - \big(E(x)\big)^2 = 1.89.$$

BINOMIAL PROBABILITY DISTRIBUTIONS

Definition: A random variable X is said to be follow binomial distribution if it assumes only non-negative values and its probability mass function is given by

$$P(X = x) = P(x) = B(x, n, p) = \begin{cases} {}^{n}C_x\, p^x q^{n-x}, & x = 0,1,2,\dots n, \quad 0 < p < 1 \text{ and } q = 1 - p. \\ 0, & otherwise \end{cases}$$

or

$$P(X = r) = {}^{n}C_r\, p^r q^{n-r} = \frac{n!}{(n-r)! \cdot r!}\, p^r q^{n-r}$$

Remarks:

1. $P(X = x) \geq 0$, $for\ all\ x \epsilon R$.
2. $\sum_{x=0}^{x=n} P(x) = \sum_{x=0}^{x=n} {}^{n}C_x\, p^x q^{n-x} = (p+q)^n = (1)^n = 1.$

Mean (Expected value) and Variance of Binomial Probability Distribution:

Derive the mean and variance of the binomial distribution.

Let X be a random variable with the Binomial distribution. The probability function of X is $P(x) = {}^{n}C_x\, p^x q^{n-x}$.

Mean (expected value) of X is $E(x) = \sum_{x=0}^{x=n} x\,P(x)$

$$E(x) = \sum_{x=0}^{x=n} x\, {}^{n}C_x\, p^x q^{n-x} = \sum_{x=0}^{x=n} x\, \frac{n!}{(n-x)! \cdot x!}\, p^x q^{n-x}$$

Remove the $x = 0$ term: since the term is 0 when $x = 0$, we start the summation from $x = 1$

$$E(x) = \sum_{x=1}^{x=n} \frac{n(n-1)!}{(n-1+1-x)! \cdot (x-1)!}\, p p^{x-1} q^{n-1+1-x}$$

$$E(x) = np \sum_{x=1}^{x=n} \frac{(n-1)!}{((n-1)-(x-1))! \cdot (x-1)!} p^{x-1} q^{(n-1)-(x-1)}$$

Put $x - 1 = y \rightarrow x = y + 1$ and put $n - 1 = m$

$$E(x) = np \sum_{y=0}^{y=m} \frac{m!}{(m-y)! \cdot y!} p^y q^{m-y} = np \sum_{y=0}^{y=m} mC_y p^y q^{m-y}$$

$$E(x) = np\,(p+q)^m \; ; but\; p + q = 1$$

$$E(x) = np\,(1)^m$$

$$E(x) = np.$$

Thus mean of binomial distribution is np.

$$\text{Variance} = E(x^2) - \big(E(x)\big)^2$$

$$E(x^2) = \sum_{x=0}^{x=n} x^2\, P(x) = \sum_{x=0}^{x=n} [x(x-1) + x]\, nC_x\, p^x q^{n-x}$$

$$E(x^2) = \sum_{x=0}^{x=n} [x(x-1)]\, nC_x\, p^x q^{n-x} + \sum_{x=0}^{x=n} [x]\, nC_x\, p^x q^{n-x}$$

$$E(x^2) = \sum_{x=0}^{x=n} [x(x-1)]\, \frac{n!}{(n-x)! \cdot x!}\, p^x q^{n-x} + np$$

$$E(x^2) = \sum_{x=0}^{x=n} [x(x-1)]\, \frac{n(n-1)(n-2)!}{(n-x)! \cdot x(x-1)(x-2)!}\, p^x q^{n-x} + np$$

$$E(x^2) = n(n-1) \sum_{x=2}^{x=n} \frac{(n-2)!}{(n-x)! \cdot (x-2)!} p^2\, p^{x-2} q^{n-x} + np$$

$$E(x^2) = n(n-1)p^2 \sum_{x=2}^{x=n} \frac{(n-2)!}{(n-2+2-x)! \cdot (x-2)!} p^{x-2} q^{n-2+2-x} + np$$

$$E(x^2) = n(n-1)p^2 \sum_{x-2=0}^{x=n} \frac{(n-2)!}{\big((n-2)-(x-2)\big)! \cdot (x-2)!} p^{x-2} q^{(n-2)-(x-2)} + np$$

Put $x - 2 = y \rightarrow x = y + 2$ and $n - 2 = m$

$$E(x^2) = n(n-1)p^2 \sum_{y=0}^{y=m} \frac{m!}{(m-y)! \cdot y!} p^y q^{m-y} + np$$

$E(x^2) = n(n-1)p^2(p+q)^m + np = n(n-1)p^2(1)^m + np = n^2p^2 - np^2 + np$

Variance $= E(x^2) - \left(E(x)\right)^2$

Variance $= n^2p^2 - np^2 + np - (np)^2 = np(1-p) = npq$

Thus Variance of binomial distribution is npq.

Examples

1. An unbiased coin is tossed 10 times. Find the probability of getting exactly 6 heads, at least 6 heads.

   Solution: Let $n = 10$ and probability of head is $p = \frac{1}{2}$ and $q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2}$

   Use Binomial Probability distribution $P(r) = nC_r\, p^r q^{n-r}$

   (i) The probability of getting exactly 6 heads is $= P(6)$

   $\therefore P(6) = 10C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^{10-6} = 0.205078.$

   (ii) The probability of getting at least 6 heads is $P(X \geq 6)$

   $\therefore P(X \geq 6) = P(6) + P(7) + P(8) + P(9) + P(10)$

   $\therefore P(X \geq 6) = 10C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^{10-6} + 10C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^{10-7} + 10C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^{10-8} +$

   $10C_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^{10-9} + 10C_{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^{10-10}$

   $\therefore P(X \geq 6) = 0.205078 + 0.117188 + 0.43945 + 0.009766 + 0.000977$

   $\therefore P(X \geq 6) = 0.376953.$

2. Probability of Man aged 60 years will live for 70 years is 1/10. Find the probability of 5 men selected at random 2 will live for 70 years.

   Solution: Let $n = 5$ and probability $p = \frac{1}{10}$ and $\therefore q = 1 - p = 1 - \frac{1}{10} = \frac{9}{10}$

   Use Binomial Probability distribution $P(r) = nC_r\, p^r q^{n-r}$

   The probability of 2 men will live for 70 years is $= P(2)$

   $\therefore P(2) = 5C_2 \left(\frac{1}{10}\right)^2 \left(\frac{9}{10}\right)^{5-2} = 0.00729.$

3. 20% bolts are produced by machine are defective. Find the probability that out of three bolts are chosen at random 1 is defective.

   Solution: Let $n = 3$ and probability $p = 20\% = \frac{20}{100} = 0.2$ and $\therefore q = 1 - p = 0.8$

Use Binomial Probability distribution $P(r) = nC_r\, p^r q^{n-r}$

The probability of 1defective bolt is $= P(1)$

$\therefore P(1) = 3C_1(0.2)^1(0.8)^2 = 0.384.$

4. The probability that hit a target in shooting practice is 0.3. If the shoots 10 times, find the probability that he hits the target.

   Solution: Let $n = 10$ and probability $p = 0.3$ and $\therefore q = 1 - p = 0.7$

   Use Binomial Probability distribution $P(r) = nC_r\, p^r q^{n-r}$

   Hit the target at $r = 1, 2, \ldots 10$

   Hence not hit the target at $r = 0$

   Hence the probability that he hits the target is $= P(X \geq 1)$

   $\therefore P(X \geq 1) = 1 - P(0)$

   $\therefore P(X \geq 1) = 1 - 10C_0(0.3)^0(0.7)^{10-0}$

   $\therefore P(X \geq 1) = 1 - 0.028248 = 0.971752.$

5. A box contains 100 bulbs out of which 10 are defective. A sample of 5 bulbs is drawn. Find the probability that none is defective.

   Solution: Let $n = 5$ and probability $p = \frac{10}{100} = 0.1$ and $\therefore q = 1 - p = 0.9$

   Use Binomial Probability distribution $P(r) = nC_r\, p^r q^{n-r}$

   Hence the probability that no defective bulb is $= P(0)$

   $\therefore P(0) = 10C_0(0.1)^0(0.9)^{10-0} = 0.348678.$

6. Out of 2000 families with 4 children each, find the number of families you would expect to have (i) at least one boy. (ii) two boys. (iii) no girl.

   Solution: Let $n = 4$, probability of boy $p = \frac{1}{2} = 0.5$ and probability of girl $q = 0.5$

   At least one boy $(r = 1, 2, 3, 4)$

   Two boys $(r = 2)$

   One or two girls $(r = 1\ and\ r = 2)$

   No girl $(r = 0)$

   Use Binomial Probability distribution $P(r) = nC_r\, p^r q^{n-r}$

   (i) The probability that at least one boy is $P(X \geq 1) = P(1) + P(2) + P(3) + P(4)$ or $P(X \geq 1) = 1 - P(0)$

$P(X \geq 1) = 1 - 4C_0(0.5)^0(0.5)^{4-0}$

$P(X \geq 1) = 0.9375.$

So, the number of families you would expect to have at least one boy is

$= 2000 \times P(X \geq 1)$

$= 2000 \times 0.9375 = 1875$

(ii) The probability of two boys is $P(2) = 4C_2(0.5)^2(0.5)^{4-2} = 4C_2(0.5)^4$

$P(2) = 0.75.$

So, the number of families you would expect to have two boys is $= 2000 \times P(2)$

$= 2000 \times 0.75 = 1500$

(iii) The probability of no girls is $P(0)$

$P(0) = 4C_0(0.5)^0(0.5)^{4-0} = 0.0625$

So, the number of families you would expect to have no girl is $= 2000 \times P(0)$

$= 2000 \times 0.0625 = 125.$

7. The mean and variance of Binomial probability distribution are 6 and 2 resp. Find $p(r \geqslant 2)$.

Solution: Use Binomial Probability distribution $P(r) = nC_r\, p^r q^{n-r}$

Given that Mean=6, and Variance=2 but we know that $Mean = np$

$Variance = npq$ hence $6 = np, and\ 2 = npq$

After solving $2 = 6q \rightarrow q = \frac{1}{3}$ and consider $p + q = 1 \rightarrow p = 1 - q = \frac{2}{3}$

Now $6 = np \rightarrow 6 = n\frac{2}{3} \rightarrow n = 9$ and $r = 0,1, \cdots 9.$

Hence $P(r \geqslant 2) = P(2) + P(3) + \cdots + P(9)$ or

$P(r \geqslant 2) = 1 - [P(0) + P(1)]$

$$P(r \geqslant 2) = 1 - \left[{}^9C_0 \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^9 + {}^9C_1 \left(\frac{2}{3}\right)^1 \left(\frac{1}{3}\right)^{9-1} \right]$$

$P(r \geqslant 2) = 1 - [0.0000508053 + 0.000914495]$

$P(r \geqslant 2) = 1 - [0.0009653]$

$P(r \geqslant 2) = 0.9990347$

8. An insurance agent accepts policies of 5 men of identical age and in good health. The probability that a man of this age will live 30 years hence is 2/3. Find the probability that in 30 years (i) all five men and (ii) at least one man will be alive.

   Solution: Use Binomial Probability distribution $P(r) = nC_r\, p^r q^{n-r}$

   Let X be the number of men of identical age and in good health those will live 30 years hence among those 5. i.e. $X \sim B\left(n = 5, p = \frac{2}{3}\right)$

   The probability mass function of X is given by $P(X = r) = {}^5C_r \left(\frac{2}{3}\right)^r \left(\frac{1}{3}\right)^{5-r}$

   where $r = 0,1, \cdots 5$

   (i) The probability that all five men will be live 30 years hence is $P(X = 5) = P(5)$

   $P(5) = {}^5C_5 \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)^0 = 0.1317$

   (ii) The probability that at least one man will be live 30 years hence is $P(X \geq 1)$

   $P(X \geq 1) = 1 - P(X = 0)$

   $P(X \geq 1) = 1 - {}^5C_0 \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^5$

   $P(X \geq 1) = 1 - 0.0041$

   $P(X \geq 1) = 0.9959.$

9. On an average, a box containing 10 articles is likely to have 2 defectives. If we consider a consignment of 100 boxes,

   (i) how many of them are expected to have at the most one defective?

   (ii) how many of them are expected to have three or less defective?

   Solution: Use Binomial Probability distribution $P(r) = nC_r\, p^r q^{n-r}$

   Let X be the number of defective articles in a box containing 10 articles.

   And $\left(n = 10, p = \frac{2}{10} = 0.2\right)$

   The probability mass function of X is given by $P(X = r) = {}^{10}C_r (0.2)^r (0.8)^{10-r}$,

   Where $r = 0,1, \cdots, 10$

   (i)The probability that a box containing 10 articles has at the most one defective is

   $P(X \leq 1) = P(X = 0) + P(X = 1)$

$P(X \leq 1) = {}^{10}C_0(0.2)^0(0.8)^{10} + {}^{10}C_1(0.2)^1(0.8)^9$

$P(X \leq 1) = 0.1074 + 0.2684$

$P(X \leq 1) = 0.3758$

So, the expected number of boxes from a consignment having at the most one defective box is $= 100 \times P(X \leq 1) = 100 \times 0.3758 = 37.58 \approx 38$.

(ii)The probability that a box containing 10 articles has at three or less defective is

$P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$

$P(X \leq 3) = 10C_0(0.2)^0(0.8)^{10} + 10C_1(0.2)^1(0.8)^9 + 10C_2(0.2)^2(0.8)^8 + 10C_3(0.2)^3(0.8)^7$

$P(X \leq 3) = 0.1074 + 0.2684 + 0.30199 + 0.201327$

$P(X \leq 3) = 0.879117$

So, the expected number of boxes from a consignment having at three or less defective boxes is $= 100 \times P(X \leq 3) = 100 \times 0.879117 = 87.91 \approx 88$.

10. On an average 20% of the workers in an industry suffer with a certain disease. If 12 workers are chosen from the industry, find the probability that (i) Exactly 2 workers suffer from the disease, (ii) At least one worker suffers from the disease.

Solution: Use Binomial Probability distribution $P(r) = nC_r\, p^r q^{n-r}$

Let X be the number of workers who suffer with the certain disease out of 12 chosen from the industry. And $(n = 12, p = 0.2)$

The probability mass function of X is given by

$P(X = r) = {}^{12}C_r(0.2)^r(0.8)^{12-r}; r = 0,1,\cdots,12$

(i) The probability that exactly 2 workers suffer from the disease is $P(X = 2)$

$P(X = 2) = {}^{12}C_2(0.2)^2(0.8)^{10} = 0.2835$

(ii) The probability that at least 1 worker suffers from the disease is $P(X \geq 1)$

$P(X \geq 1) = 1 - P(X = 0)$

$P(X \geq 1) = 1 - 0.0687$

$P(X \geq 1) = 0.9313$.

11. A coffee connoisseur claims that he can distinguish between a cup of instant coffee and a cup percolator coffee 75% of the time. It is agreed that his claim will be accepted if he

correctly identifies at least 5 of the 6 cups. Find the chances of having the claim (i) accepted (ii) rejected, when he does have the ability of claims.

Solution*:* Use Binomial Probability distribution $P(r) = nC_r\, p^r q^{n-r}$

Let n=6 (total number of trials/cups).

$p = \frac{75}{100} = 0.75$ (probability of a correct identification).

$q = 1 - p = 0.25$ (probability of an incorrect identification).

(i) Accepted: Required Probability $P(r \geq 5) = P(5) + P(6) = 0.355 + 0.177 = 0.534$

(ii) Rejected: Required Probability $1 - P(r \geq 5) = 0.466$

12. A series of five one day matches is to be played between India and Pakistan. Assuming that the probability of India's win in each match as 0.6 and results of all the five matches independent of each other, find the probability that (i) India wins at least one match, (ii) India wins the series.

Solution: Use Binomial Probability distribution $P(r) = nC_r\, p^r q^{n-r}$

Let X be the number of matches out of 5 those will be won by India. and $(n = 5, p = 0.6)$

The probability mass function of X is given by $P(X = r) = {}^5C_r(0.6)^r(0.4)^{5-r}$;

Where $r = 0,1,2,3,4,5$.

(i) The probability that India wins at least one match is $= P(X \geq 1)$

$P(X \geq 1) = 1 - P(X = 0)$

$P(X \geq 1) = 1 - {}^5C_0(0.6)^0(0.4)^5$

$P(X \geq 1) = 1 - 0.01024$

$P(X \geq 1) = 0.98976.$

(ii) The probability that India wins the series is $P(X \geq 3)$

$P(X \geq 3) = P(X = 3) + P(X = 4) + P(X = 5)$

$P(X \geq 3) = {}^5C_3(0.6)^3(0.4)^2 + {}^5C_4(0.6)^4(0.4)^1 + {}^5C_5(0.6)^5(0.4)^0$

$P(X \geq 3) = 0.3456 + 0.2592 + 0.0778$

$P(X \geq 3) = 0.6826.$

13. In a binomial probability distribution consisting of 5 independent trials, probabilities of 1 and 2 successes are found to be 0.4096 and 0.2048 respectively. Find the parameter p of the distribution.

Solution: Use Binomial Probability distribution $P(r) = nC_r\, p^r q^{n-r}$ and $p + q = 1$

Given that $n = 5$, $P(X = 1) = 0.4096$ and $P(X = 2) = 0.2048$ that is

$P(1) = 5C_1\, p^1(1-p)^{5-1} \rightarrow 5p(1-p)^4 = 0.4096 \quad \ldots i$

$P(2) = 5C_2\, p^2(1-p)^{5-2} \rightarrow 10p^2\,(1-p)^3 = 0.2048 \quad \ldots ii$

By division of equations (i) and (ii)

$$\frac{5p(1-p)^4}{10p^2\,(1-p)^3} = \frac{0.4096}{0.2048}$$

$$\frac{1-p}{2p} = 2 \rightarrow\rightarrow \frac{1}{p} - 1 = 4 \rightarrow\rightarrow \frac{1}{p} = 5 \rightarrow\rightarrow p = \frac{1}{5}$$

$p = 0.2$.

14. With the usual notations, find the probability of success p for a binomial distribution, if $n = 6$, and $9P(x = 4) = P(x = 2)$.

Solution: Use Binomial Probability distribution $P(r) = nC_r\, p^r q^{n-r}$ and $p + q = 1$

Consider $9P(x = 4) = P(x = 2)$ and use $q = 1 - p$

$9\ 6C_4\, p^4(1-p)^{6-4} = 6C_2\, p^2(1-p)^{6-2}$

$$9\frac{6!}{2!\,4!}\, p^4(1-p)^2 = \frac{6!}{2!\,4!}\, p^2(1-p)^4$$

$9p^2 = (1-p)^2$

$9p^2 - p^2 + 2p - 1 = 0$

$8p^2 + 2p - 1 = 0$

$$p = -\frac{1}{2}, \frac{1}{4}$$

Hence $p = \frac{1}{4}$. $\left(p = -\frac{1}{2} is\ rejected, Since\ probability\ cannot\ be\ negative\right)$.

POISSON PROBABILITY DISTRIBUTION

A discrete random variable X is said to follow a Poisson distribution with parameter z if its probability mass function is given by,

$$P(X = r) = \frac{e^{-z}z^r}{r!} : z > 0, r = 0,1, \cdots$$

Where z is Average number of events in the interval $OR\ \ z = np$

Also n -number of trials is very large, p-probability of success at each trial.

Remarks:

1. $P(X = x) \geq 0$, $for\ all\ x \epsilon R$.
2. $\sum_{x=0}^{x=n} P(x) = \sum_{x=0}^{x=\infty} \frac{e^{-z} z^x}{x!} = e^{-z} \sum_{x=0}^{x=\infty} \frac{z^x}{x!} = e^{-z}\left[1 + \frac{z}{1!} + \frac{z^2}{2!} + \cdots\right] = e^{-z}[e^z] = 1.$

Mean (Expected value) of Poisson Probability Distribution:

Derive the mean and variance of the Poisson distribution.

Let X be a random variable with the Poisson distribution. The probability function of X is

$P(x) = \frac{e^{-z} z^x}{x!}$.

Mean (expected value) of X is $E(x) = \sum_{x=0}^{x=\infty} x\, P(x)$

$$E(x) = \sum_{x=0}^{x=\infty} x \frac{e^{-z} z^x}{x!}$$

$$E(x) = e^{-z} \sum_{x=0}^{x=n} \frac{x\, z^x}{x!}$$

$$E(x) = e^{-z}\left[0 + \frac{z}{1!} + \frac{2z^2}{2!} + \frac{3z^3}{3!} + \frac{4z^4}{4!} + \cdots\right]$$

$$E(x) = e^{-z} z\left[1 + \frac{z}{1!} + \frac{3z^2}{3\,(2!)} + \frac{4z^3}{4\,(3!)} + \frac{5z^4}{5\,(4!)} + \cdots\right]$$

$$E(x) = z\, e^{-z}\left[1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \cdots\right]$$

Use $1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \cdots = e^z$

Mean $= E(x) = z e^{-z}\, e^z$.

Mean $= E(x) = z$.

Thus Mean of poisson distribution is z.

Variance $= E(x^2) - \left(E(x)\right)^2$

$$E(x^2) = \sum_{x=0}^{x=\infty} x^2\, P(x) = \sum_{x=0}^{x=\infty} [x^2 - x + x]\, P(x) = \sum_{x=0}^{x=\infty} [x(x-1) + x]\, P(x)$$

$$E(x^2) = \sum_{x=0}^{x=\infty} [x(x-1)]\, P(x) + \sum_{x=0}^{x=\infty} xP(x)$$

$$E(x^2) = \sum_{x=0}^{x=\infty} [x(x-1)]\, \frac{e^{-z} z^x}{x!} + z$$

$$E(x^2) = e^{-z} \sum_{x=0}^{x=\infty} [(x^2 - x)]\, \frac{z^x}{x!} + z$$

$$E(x^2) = e^{-z} \left[0 + 0 + \frac{2z^2}{2!} + \frac{6z^3}{3!} + \frac{12z^4}{4!} + \frac{20z^5}{5!} + \frac{30z^6}{6!} + \cdots\right] + z$$

$$E(x^2) = e^{-z} z^2 \left[1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \cdots\right] + z$$

Use $1 + \frac{z}{1!} + \frac{z^2}{2!} + \cdots += e^z$

$$E(x^2) = z^2 e^{-z} e^z + z$$

$$E(x^2) = z^2 + z$$

$$\text{Variance} = E(x^2) - \big(E(x)\big)^2$$

$$\text{Variance} = z^2 + z - z^2$$

$$\text{Variance} = z.$$

Thus Variance of poisson distribution is z.

Examples

1. In a certain factory turning out razor blades, there is a small chance of $1/500$ for any blade to be defective. The blades are supplied in a packet of 10. Use Poisson distribution to calculate number of packets containing no defective and two defective blades, in a consignment of 10000 packets.

   Solution: Let $n = 1000$, probability of defective blades is $p = \frac{1}{500}$ and

   mean $z = np = 10 * \frac{1}{500} = 0.02$

   Use Poisson Probability Distribution: $P(r) = \frac{e^{-z} z^r}{r!}$

(i) The probability of no defective blade is $= P(0)$

$$P(0) = \frac{e^{-z}z^0}{0!} = e^{-z} = e^{-0.02} = 0.980199$$

So, the number of packets containing no defective blades, in a consignment of 10000 packets is $= 10000 \times P(0) = 10000 \times e^{-0.02} = 9801.99 \approx 9802.$

(ii) The probability of two defective blade is $P(2)$

$$P(2) = \frac{e^{-z}z^2}{2!} = \frac{e^{-0.02}0.02^2}{2} = 0.000196$$

So, the number of packets containing two defective blades, in a consignment of 10000 packets is $= 10000 \times P(2) = 1.960397 \approx 2.$

2. The average number of misprints per page of a book is 1.5. Assuming the distribution of number of misprints to be Poisson. Find (i) the prob that a particular book is free from misprints. (ii) Number of pages containing more than one misprint if the book contains 900 pages.

Solution: Given that, mean $z = 1.5$

Use Poisson Probability Distribution: $P(r) = \frac{e^{-z}z^r}{r!}$

(i) The probability that a particular book is free from misprints is $= P(0)$

$$P(0) = \frac{e^{-z}z^0}{0!} = e^{-z} = e^{-1.5} = 0.22313$$

(ii) A particular book containing more than one misprint at $r = 2, 3, 4, \ldots$ and hence the probability is $= P(X > 1) = P(2) + P(3) + P(4) + \cdots$ OR

$$P(X > 1) = 1 - [P(0) + P(1)]$$

$$P(X > 1) = 1 - \left[\frac{e^{-z}z^0}{0!} + \frac{e^{-z}z^1}{1!}\right]$$

$$P(X > 1) = 1 - [e^{-z} + e^{-z}z]$$

$$P(X > 1) = 1 - e^{-z}[1 + z]$$

$$P(X > 1) = 1 - e^{-1.5}[1 + 1.5]$$

$$P(X > 1) = 1 - 2.5e^{-1.5} = 1 - 0.557825 = 0.442175$$

So, the number of pages containing more than one misprint if the book contains 900 pages is $= 900 \times P(X > 1) = 900 \times 0.442175 = 397.95 \approx 398.$

3. Number of road accidents on a highway during a month follows a Poisson distribution with mean 5. Find the probability that in a certain month number of accidents on the highway will be (i) less than 3. (ii) Between 3 & 5. (iii) more than 3.

Solution: Given that, mean $z = 5$

Use Poisson Probability Distribution: $P(r) = \frac{e^{-z}z^r}{r!}$

(i) The probability that in a certain month number of accidents on the highway will be less than three is $= P(X < 3)$

$P(X < 3) = P(0) + P(1) + P(2)$

$$P(X < 3) = \frac{e^{-z}z^0}{0!} + \frac{e^{-z}z^1}{1!} + \frac{e^{-z}z^2}{2!}$$

$$P(X < 3) = \frac{e^{-5}5^0}{0!} + \frac{e^{-5}5^1}{1!} + \frac{e^{-5}5^2}{2!}$$

$P(X < 3) = 0.006378 + 0.03369 + 0.084224 = 0.124652.$

(ii) The probability that in a certain month number of accidents on the highway will be between three and five is $= P(3 \le X \le 5) = P(3) + P(4) + P(5)$

$$P(3 \le X \le 5) = \frac{e^{-z}z^3}{3!} + \frac{e^{-z}z^4}{4!} + \frac{e^{-z}z^5}{5!}$$

$$P(3 \le X \le 5) = \frac{e^{-5}5^3}{6} + \frac{e^{-5}5^4}{24} + \frac{e^{-5}5^5}{120} = 0.140374 + 0.175467 + 0.175467$$

$P(3 \le X \le 5) = 0.491309$

(iii) The probability that in a certain month number of accidents on the highway will be more than three is $= P(X > 3) = P(4) + P(5) + \cdots$ OR

$P(X > 3) = 1 - [P(0) + P(1) + P(2) + P(3)]$

$$P(X > 3) = 1 - \left[\frac{e^{-z}z^0}{0!} + \frac{e^{-z}z^1}{1!} + \frac{e^{-z}z^2}{2!} + \frac{e^{-z}z^3}{3!}\right]$$

$$P(X > 3) = 1 - \left[\frac{e^{-5}5^0}{1} + \frac{e^{-5}5^1}{1} + \frac{e^{-5}5^2}{2} + \frac{e^{-5}5^3}{6}\right]$$

$P(X > 3) = 1 - [0.006378 + 0.03369 + 0.084224 + 0.140374]$

$P(X > 3) = 1 - 0.265026 = 0.734974.$

4. A manufacturer of cotter pins knows that 2% of his product is defective. If he sells cotter pins in boxes of 100 pins and guarantees that not more than 5 pins will be defective in a box, find the app probability that a box will fail to meet the guaranteed quality.

   Solution: Let $n = 100$, probability of defective cotter pins is $p = 2\% = \frac{2}{100}$ and mean $z = np = 100 * \frac{2}{100} = 2$

   Use Poisson Probability Distribution: $P(r) = \frac{e^{-z}z^r}{r!}$

   Guarantee: not more than 5 pins will be defective in a box, i.e. $r = 0,1,2,3,4,5$

   Guarantee Fail: more than 5 pins will be defective in a box, i.e. $r = 6,7,8, \ldots$

   Hence the probability that a box will fail to meet the guaranteed quality is $=$

   $P(X > 5) = P(6) + P(7) + P(8) + \cdots$ OR

   $P(X > 5) = 1 - [P(0) + P(1) + P(2) + P(3) + P(4) + P(5)]$

   $$P(X > 5) = 1 - \left[\frac{e^{-z}z^0}{0!} + \frac{e^{-z}z^1}{1!} + \frac{e^{-z}z^2}{2!} + \frac{e^{-z}z^3}{3!} + \frac{e^{-z}z^4}{4!} + \frac{e^{-z}z^5}{5!}\right]$$

   $$P(X > 5) = 1 - \left[\frac{e^{-2}2^0}{0!} + \frac{e^{-2}2^1}{1!} + \frac{e^{-2}2^2}{2!} + \frac{e^{-2}2^3}{3!} + \frac{e^{-2}2^4}{4!} + \frac{e^{-2}2^5}{5!}\right]$$

   $P(X > 5) = 1 - [0.13533 + 0.2706 + 0.2706 + 0.18044 + 0.09022 + 0.03608]$

   $P(X > 5) = 1 - [0.983436]$

   $P(X > 5) = 0.016564.$

5. Between 2 pm and 3 pm, the average numbers of phone calls per minute coming into the company are 2. Find the probability that during one particular minute, there will be [i] no phone call at all. [ii] 2 or less calls.

   Solution: Given that, mean $z = 2$,

   Use Poisson Probability Distribution: $P(r) = \frac{e^{-z}z^r}{r!}$

   (i) The probability that during one particular minute, there will be no phone call is

   $$P(0) = \frac{e^{-z}z^0}{0!} = e^{-2} = 0.135335$$

   (ii) The probability that during one particular minute, there will be 2 or less calls is

$$P(X \leq 2) = P(0) + P(1) + P(2) = \frac{e^{-z}z^0}{0!} + \frac{e^{-z}z^1}{1!} + \frac{e^{-z}z^2}{2!}$$

$$P(X \leq 2) = \frac{e^{-z}2^0}{0!} + \frac{e^{-z}2^1}{1!} + \frac{e^{-z}2^2}{2!} = 0.135335 + 0.270671 + 0.270671$$

$$P(X \leq 2) = 0.676676.$$

6. If the probability that a concrete cube fails is 0.001. Determine the probability that out of 1000 cubes: [i] Exactly two [ii] More than one cube will fail.

Solution: Let $n = 1000$, probability of failed concrete cubes is $p = 0.001$ and mean $z = np = 1000 * 0.001 = 1$

Use Poisson Probability Distribution: $P(r) = \frac{e^{-z}z^r}{r!}$

(i) The probability that out of 1000 cubes, exactly two cubes will fail is $P(2)$

$$P(2) = \frac{e^{-z}z^2}{2!} = \frac{e^{-1}1^2}{2} = 0.18394.$$

(ii) The probability that out of 1000 cubes, more than one cube will fail is $P(X > 1)$

OR $P(X > 1) = 1 - P(0)$

$$P(X > 1) = 1 - \frac{e^{-z}z^0}{0!} = 1 - e^{-1} = 1 - 0.367879 = 0.632121.$$

7. There is a small probability of 1/1,000 for any computer produced to be defective. Determine in a sample of 2,000 computers, the probability that there (i) is no defective and (ii) are two defectives.

Solution: Let $n = 2000, p = \frac{1}{1000}$ *hence* mean is $z = np = 2000 \times \frac{1}{1000} = 2$

Use Poisson Probability Distribution: $P(r) = \frac{e^{-z}z^r}{r!}$

(i) The probability that there is no defective computer is $P(0)$

$$P(0) = \frac{e^{-2}2^0}{0!} = e^{-2} = 0.1353.$$

(ii) The probability that there are two defective computers is $P(2)$

$$P(2) = \frac{e^{-2}2^2}{2!} = 0.2706.$$